

Customer Behavior Analysis

End-to-End Data Analytics Project | MySQL · Python · K-Means

Project by: Student Name | Final Semester Project | 2024

Project Overview

Objective

Analyze customer purchasing patterns using SQL + Python to extract actionable business insights.

Data Source

MySQL relational database with 4 tables: Customers, Products, Orders, Order Items.

Tools Used

MySQL · Python · Pandas · Matplotlib · Seaborn · Scikit-learn (K-Means)

Output

8 visualizations + customer segmentation + key insights report.

Project Pipeline



Raw MySQL data → CSV → Pandas cleaning → Analysis → 8 Charts + K-Means Segmentation → Insights

Database Schema — 4 Tables

customers

- ▶ customer_id (PK)
- ▶ name
- ▶ city
- ▶ signup_date

products

- ▶ product_id (PK)
- ▶ product_name
- ▶ category
- ▶ price

orders

- ▶ order_id (PK)
- ▶ customer_id (FK)
- ▶ order_date

order_items

- ▶ order_item_id (PK)
- ▶ order_id (FK)
- ▶ product_id (FK)
- ▶ quantity

customers — orders — order_items — products

Dataset at a Glance

15

Customers

13

Products

20

Orders

22

Order Items

Cities Covered

Ahmedabad • Mumbai • Delhi • Pune • Bangalore
Hyderabad • Jaipur • Lucknow • Chennai

Product Categories

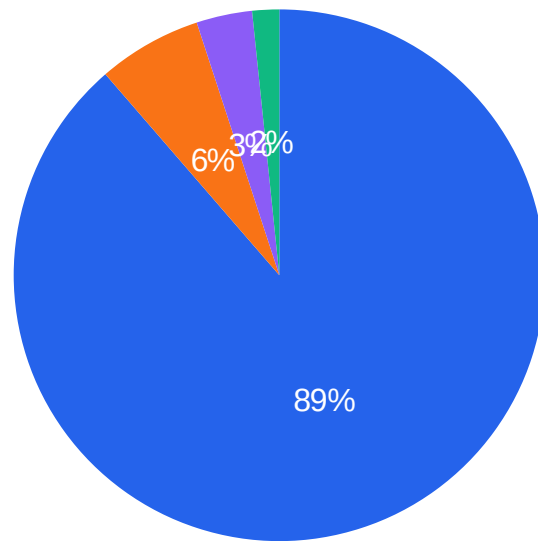
-  Electronics — High Revenue
-  Fashion — Medium Revenue
-  Accessories — Medium Revenue
-  Furniture — Niche

Revenue Analysis

Monthly Revenue Trend



Revenue by Category



Rs.5.42L

Total Revenue

Rs.27.1K

Avg Order Value

July

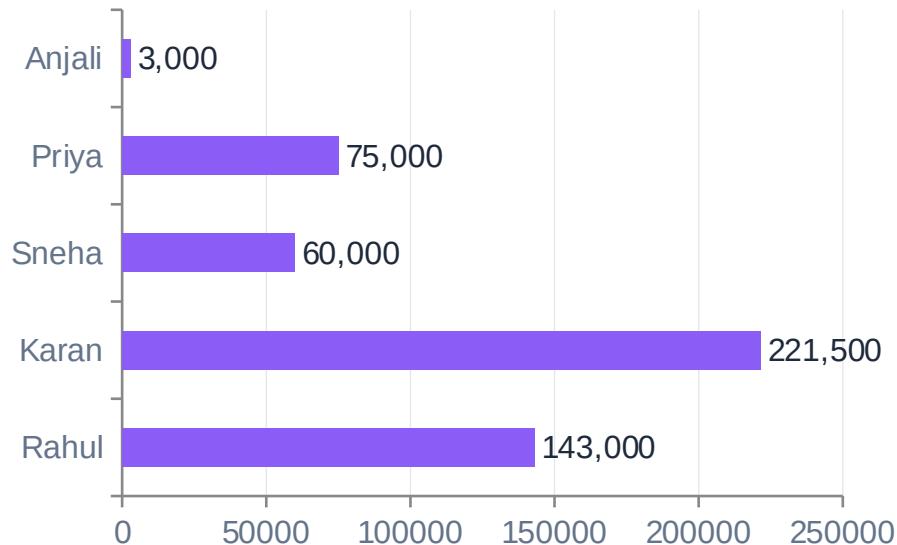
Peak Month

70%

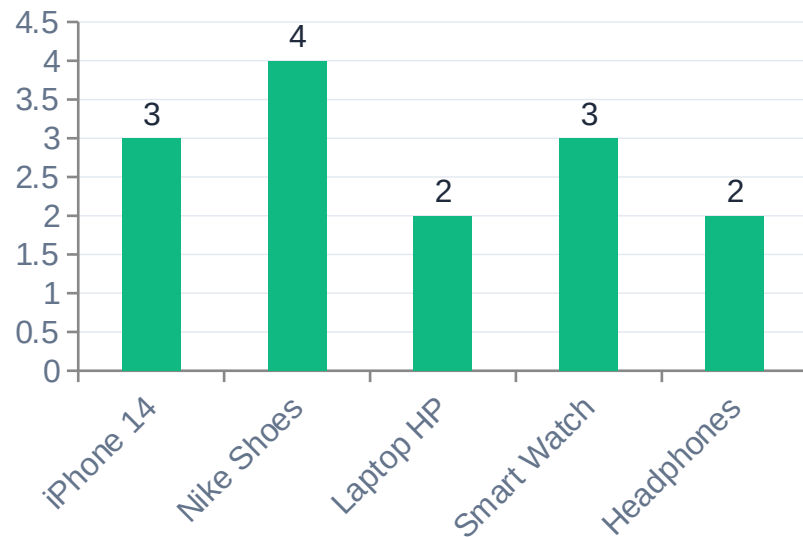
Electronics Share

Top Customers & Products

Top 5 Customers by Revenue



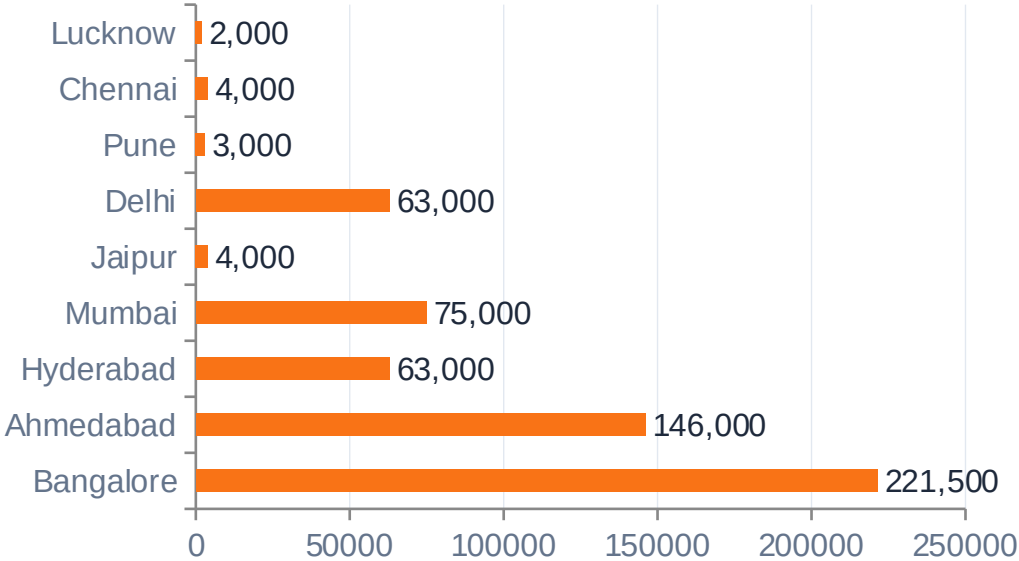
Top Products by Units Sold



Karan has highest total spend across 2 orders | iPhone 14 & Nike Shoes are most popular products

City-wise Revenue & Heatmap

Revenue by City



Category × City Heatmap			
Category	Ahm.	Blr.	Mum.
Electronics	143K	222K	70K
Fashion	0	6K	10K
Accessories	3K	0	0
Furniture	0	0	0



Bangalore & Ahmedabad drive majority of Electronics revenue

Customer Segmentation — K-Means (3 Clusters)

High-Value

4 customers

Rs.1.10L avg

Premium buyers — Electronics focus.
Highest lifetime value.

Medium-Value

6 customers

Rs.14K avg

Regular buyers — Mixed categories.
Growth potential.

Low-Value

5 customers

Rs.3K avg

Occasional buyers —
Fashion/Accessories. Need re-
engagement.

Algorithm: K-Means Clustering (k=3) | Features: Total Revenue, Total Quantity, Total Orders | Scaling: StandardScaler

SQL Techniques Used



JOINS

INNER JOIN across 4 tables to build analytical dataset.



GROUP BY / HAVING

Group by city, category, month — HAVING for repeat customers.



View

monthly_revenue_view — reusable query abstraction.



Trigger

log_new_order — auto-logs every new order insert.

Σ Aggregations

SUM, COUNT, AVG for revenue, order frequency, avg value.



Window Functions

RANK(), LAG(), SUM() OVER for ranking and running totals.



Stored Procedure

GetRevenueByCity(city) — parameterized city revenue query.



Subquery / CTE

Nested queries for monthly growth % calculation.

Key Insights & Findings



Total Revenue: Rs.5.42 Lakhs

Generated from 20 orders across 15 customers over 9 months (Jan-Sep 2024).



Electronics Dominates (70%)

iPhone 14 and Laptop HP are the highest revenue-generating products.



Bangalore = Top City

Bangalore alone contributed Rs.2.2L due to high-value repeat orders by Karan.



Repeat Customers = 30%

Rahul, Priya, Karan, Sneha, Amit placed multiple orders — high retention signal.



Peak Month: July 2024

Highest monthly revenue of Rs.1.4L — driven by Electronics purchases.



4 High-Value Customers

K-Means identified top 4 customers contributing 65%+ of total revenue.

Technology Stack



MySQL

Database design, SQL queries, Views, Procedures, Triggers



Python

Core programming language for data processing and analysis



Pandas

Data loading, cleaning, transformation and aggregation



Matplotlib

Line charts, bar charts, pie charts — 8 visualizations created



Seaborn

Statistical heatmaps and advanced chart styling



Scikit-learn

K-Means clustering for customer segmentation into 3 groups

Conclusion

- ✓ Built a complete end-to-end Customer Behavior Analysis system from scratch.
- ✓ Designed a normalized MySQL database with 4 related tables and 20+ records.
- ✓ Wrote complex SQL — Joins, Aggregations, Window Functions, Views, Procedures, Triggers.
- ✓ Exported data to CSV and performed full EDA using Pandas, Matplotlib and Seaborn.
- ✓ Identified top customers, peak months, best categories and cities by revenue.
- ✓ Applied K-Means clustering to segment customers into High, Medium and Low value groups.

Resume: Built end-to-end Customer Behavior Analysis using MySQL, Python, Pandas, Matplotlib & K-Means clustering.